

Probability and Statistics Notes

Prob Stats 1: Notes, Exercises and PYQs

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Part I

Probability

Chapter 1

Continuous Random Variables and Distributions

1.1 Continuous Random Variables

Let us first define what a continuous function and a continuous random variable is.

Definition 1.1.0.1

Continuous Function

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be continuous at a point $x_0 \in \mathbb{R}$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x \in \mathbb{R}$, if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \epsilon$. If a function is continuous at every point in its domain, it is called a continuous function.

Let us consider a simple example of a continuous function ($f(x) = \sin(x)$) -

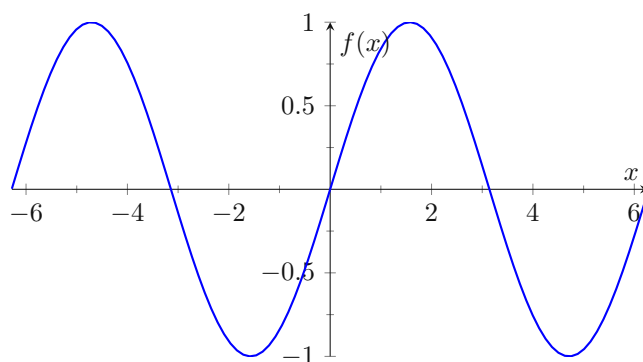


Figure 1.1: Graph of the continuous function $f(x) = \sin(x)$

Definition 1.1.0.2

Continuous Random Variable

A random variable X is said to be continuous if its cumulative distribution function (CDF) $F_X(x) = P(X \leq x)$ is continuous for all $x \in \mathbb{R}$.

Now, we define the probability density function (PDF) for a continuous random variable.

Definition 1.1.0.3

Probability Density Function (PDF)

The probability density function (PDF) of a continuous random variable X is a function $f_X(x)$

such that for any two numbers a and b with $a < b$,

$$P(a < X < b) = \int_a^b f_X(x) dx$$

The PDF must satisfy the following properties:

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$
2. $\int_{-\infty}^{\infty} f_X(x) dx = 1$

We also now define the cumulative distribution function (CDF) for a continuous random variable.

Definition 1.1.0.4

Cumulative Distribution Function (CDF)

The cumulative distribution function (CDF) of a continuous random variable X is a function $F_X(x)$ defined as

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

where $f_X(t)$ is the probability density function (PDF) of X .

Now, we are equipped to study these continuous random variables and their distributions in detail.

Let $X \sim f(x) \wedge F(x)$, let $Y = g(X)$, where g is strictly monotonic and differentiable. Then,

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = \begin{cases} P(X \leq g^{-1}(y)), & \text{if } g \text{ is increasing} \\ P(X \geq g^{-1}(y)), & \text{if } g \text{ is decreasing} \end{cases}$$

We also have -

$$f_Y(y) = \begin{cases} f(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y), & \text{if } g \text{ is increasing} \\ f(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|, & \text{if } g \text{ is decreasing} \end{cases}$$

Therefore, we can simply write -

$$f_Y(y) = f(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right| \sim f(x) \cdot \left| \frac{dx}{dy} \right|$$

Now, let us consider the case where g is monotone, but not strictly monotone. This means that there exists an interval where g is constant. Let us denote this interval as $[a, b]$. In this case, for any y in the range of g corresponding to the interval $[a, b]$, we have -

$$P(Y = y) = P(g(X) = y) = P(X \in [a, b]) = \int_a^b f_X(x) dx$$

Let $g^{-1}(y) = \inf\{x : g(x) \geq y\}$, then we have -

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

Differentiating both sides with respect to y , we get -

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

However, since g is not strictly monotone, we need to account for the intervals where g is constant. Therefore, we can write the PDF of Y as -

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right| + \sum_i P(X \in [a_i, b_i]) \cdot \delta(y - g(c_i))$$

where $[a_i, b_i]$ are the intervals where g is constant, c_i is any point in the interval $[a_i, b_i]$, and δ is the Dirac delta function. This accounts for the probability mass at the points where g is constant.

Example

Let $X \sim f(x) = \begin{cases} \frac{1}{2}, & -0.5 \leq x \leq 1.5 \\ 0, & \text{otherwise} \end{cases}$ and $Y = X^2$. We need to find the distribution of Y .

We have -

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$$

Therefore,

$$F_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x) dx = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \frac{\sqrt{y} - (-\sqrt{y})}{2} = \sqrt{y}$$

for $0 \leq y \leq 2.25$. Differentiating, we get -

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{2\sqrt{y}}$$

for $0 < y \leq 2.25$. Thus, the distribution of Y is given by -

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y \leq 2.25 \\ 0, & \text{otherwise} \end{cases}$$

1.1.1 Ludeberg Levy CLT

Theorem 1.1. *Ludeberg Levy CLT Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with mean μ and variance σ^2 . Let $S_n = X_1 + X_2 + \dots + X_n$. Then, as $n \rightarrow \infty$,*

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1)$$

where $N(0, 1)$ is the standard normal distribution.

Proof. Write $y_i = \frac{X_i - \mu}{\sigma}$, then y_i are i.i.d. with mean 0 and variance 1 for all i . Let $M(\cdot)$ be the moment generating function (MGF) of y_i . Then, we have -

$$M_{S_n^*}(t) = E[e^{t \cdot \frac{S_n - n\mu}{\sigma\sqrt{n}}}] = E[e^{t \cdot \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma\sqrt{n}}}] = E[e^{t \cdot \frac{\sum_{i=1}^n y_i}{\sqrt{n}}}]$$

Since y_i are i.i.d., we can write -

$$M_{S_n^*}(t) = \prod_{i=1}^n E[e^{t \cdot \frac{y_i}{\sqrt{n}}}] = \left(M\left(\frac{t}{\sqrt{n}}\right) \right)^n$$

Now, we can use the Taylor expansion of $M(t)$ around $t = 0$ to get -

$$M\left(\frac{t}{\sqrt{n}}\right) = 1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right)$$

Therefore, we have -

$$M_{S_n^*}(t) = \left(1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right)^n$$

Taking the limit as $n \rightarrow \infty$, we get -

$$\lim_{n \rightarrow \infty} M_{S_n^*}(t) = \lim_{n \rightarrow \infty} \left(1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right)^n = e^{\frac{t^2}{2}}$$

This is the MGF of the standard normal distribution. Therefore, by the uniqueness of MGFs, we have -

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1)$$

□

Theorem 1.2 (De Moivre-Laplace CLT). *Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with mean μ and variance σ^2 . Let $S_n = X_1 + X_2 + \dots + X_n$. Then, as $n \rightarrow \infty$,*

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1)$$

where $N(0, 1)$ is the standard normal distribution.

Proof. The proof of the De Moivre-Laplace CLT is similar to the proof of the Ludeberg Levy CLT, but it uses the characteristic function instead of the moment generating function. Let $\phi(t)$ be the characteristic function of y_i . Then, we have -

$$\phi_{S_n^*}(t) = E[e^{it \cdot \frac{S_n - n\mu}{\sigma\sqrt{n}}}] = E[e^{it \cdot \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma\sqrt{n}}}] = E[e^{it \cdot \frac{\sum_{i=1}^n y_i}{\sqrt{n}}}]$$

Since y_i are i.i.d., we can write -

$$\phi_{S_n^*}(t) = \prod_{i=1}^n E[e^{it \cdot \frac{y_i}{\sqrt{n}}}] = \left(\phi\left(\frac{t}{\sqrt{n}}\right) \right)^n$$

Now, we can use the Taylor expansion of $\phi(t)$ around $t = 0$ to get -

$$\phi\left(\frac{t}{\sqrt{n}}\right) = 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)$$

Therefore, we have -

$$\phi_{S_n^*}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right)^n$$

Taking the limit as $n \rightarrow \infty$, we get -

$$\lim_{n \rightarrow \infty} \phi_{S_n^*}(t) = \lim_{n \rightarrow \infty} \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right)^n = e^{-\frac{t^2}{2}}$$

This is the characteristic function of the standard normal distribution. Therefore, by the uniqueness of characteristic functions, we have -

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1)$$

□

Part II

Statistics

Let $(x_i, y_i), i \in \{1, 2, \dots, n\}$ be n paired observations.

$$(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\implies (x, y) = \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y}$$

$$(x) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2$$

$$(y) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 = \frac{1}{n} \sum_{i=1}^n y_i^2 - \bar{y}^2$$

1.1.2 Identities of Covariance and Variance

- $(ax, by) = ab \cdot (x, y)$
- $(x, x) = (x)$
- $(ax + b, cy + d) = ac \cdot (x, y)$

Let ρ be the correlation coefficient between x and y . Then, we have -

$$\rho = \frac{(x, y)}{\sqrt{(x) \cdot (y)}} = \frac{\frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y}}{\sqrt{\left(\frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2\right) \cdot \left(\frac{1}{n} \sum_{i=1}^n y_i^2 - \bar{y}^2\right)}}$$